

HDOC - Day 12

Menu-Driven C Program Using Switch-Case

Problem Statement

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Find the sum and product of the first and last digits.
- Count how many digits are less than, equal to, and greater than a given digit k.

Algorithm

1. Start.
2. Input a positive integer num.
3. If $\text{num} \leq 0$, display "Please enter a positive integer" and stop.
4. Display the menu: 1. Sum and product of first and last digits 2. Compare digits with k.
5. Input the user's choice.
6. Use `switch(choice)`.
7. Case 1: Set $\text{last} = \text{num} \% 10$. Set $\text{temp} = \text{num}$. Repeatedly divide temp by 10 while $\text{temp} \geq 10$. The remaining value is the first digit. Calculate $\text{sum} = \text{first} + \text{last}$ and $\text{product} = \text{first} * \text{last}$. Display the results.
8. Case 2: Input digit k. If k is outside 0 to 9, display an error. Otherwise set $\text{lesser} = \text{equal} = \text{greater} = 0$ and $\text{temp} = \text{num}$. Extract each digit using $\text{temp} \% 10$. Compare it with k and increment the appropriate counter. Divide temp by 10 and repeat until temp becomes 0. Display all three counts.
9. Default: Display "Invalid choice".
10. Stop.

Test Case Table

Test	Number	Choice	k	Expected Output
1	12345	1	-	Sum = 6, Product = 5
2	12345	2	3	Lesser = 2, Equal = 1, Greater = 2
3	123	3	-	Invalid choice
4	-12	-	-	Please enter a positive integer

C Program

```
#include <stdio.h>
```

```
int main()
```

```

{
    int num, choice, temp, first, last;
    int sum, product, k, digit;
    int lesser = 0, equal = 0, greater = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    if (num <= 0)
    {
        printf("Please enter a positive integer.\n");
        return 0;
    }

    printf("\nMENU\n");
    printf("1. Sum and product of first and last digits\n");
    printf("2. Count digits lesser than, equal to and greater than k\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            last = num % 10;
            temp = num;

            while (temp >= 10)
                temp = temp / 10;

            first = temp;
            sum = first + last;
            product = first * last;

            printf("Sum = %d\n", sum);
            printf("Product = %d\n", product);
            break;

        case 2:
            printf("Enter digit k (0-9): ");
            scanf("%d", &k);

            if (k < 0 || k > 9)
            {
                printf("Please enter k between 0 and 9.\n");
                break;
            }

            temp = num;
            while (temp > 0)
            {
                digit = temp % 10;

                if (digit < k)
                    lesser++;
                else if (digit == k)
                    equal++;
                else
                    greater++;

                temp = temp / 10;
            }
    }
}

```

```

        printf("Count_lesser = %d\n", lesser);
        printf("Count_equal = %d\n", equal);
        printf("Count_greater = %d\n", greater);
        break;

    default:
        printf("Invalid choice.\n");
    }

    return 0;
}

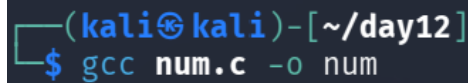
```

Compilation and Execution

Compilation command: `gcc num.c -o num`

Execution command: `./num`

Compilation

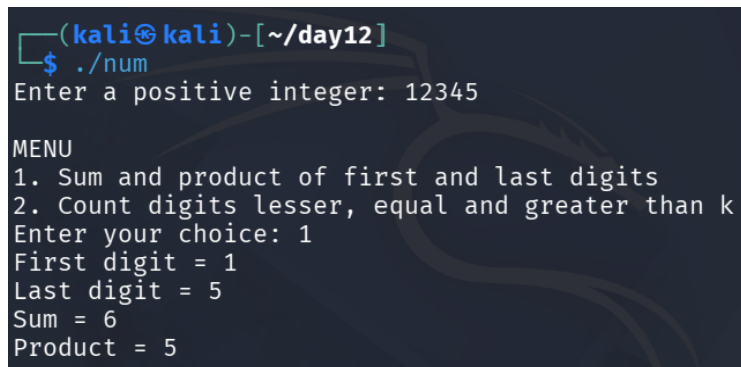


```

(kali㉿kali)-[~/day12]
$ gcc num.c -o num

```

Execution - Choice 1



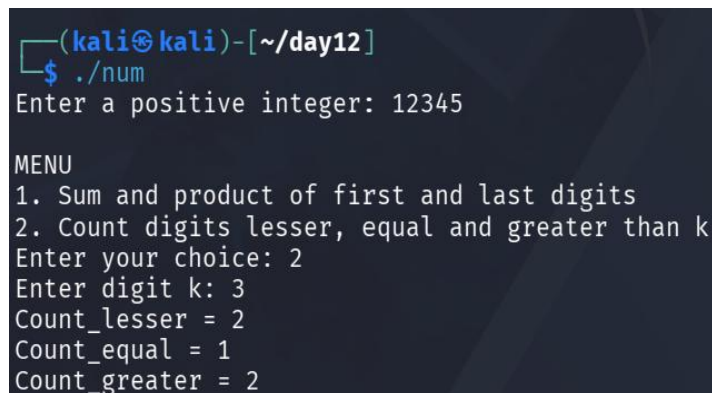
```

(kali㉿kali)-[~/day12]
$ ./num
Enter a positive integer: 12345

MENU
1. Sum and product of first and last digits
2. Count digits lesser, equal and greater than k
Enter your choice: 1
First digit = 1
Last digit = 5
Sum = 6
Product = 5

```

Execution - Choice 2



```

(kali㉿kali)-[~/day12]
$ ./num
Enter a positive integer: 12345

MENU
1. Sum and product of first and last digits
2. Count digits lesser, equal and greater than k
Enter your choice: 2
Enter digit k: 3
Count_lesser = 2
Count_equal = 1
Count_greater = 2

```

Execution - Invalid Choice

```
(kali@kali)-[~/day12]
$ ./num
Enter a positive integer: 123

MENU
1. Sum and product of first and last digits
2. Count digits lesser, equal and greater than k
Enter your choice: 3
Invalid choice.
```

Execution - Invalid Positive Integer

```
(kali@kali)-[~/day12]
$ ./num
Enter a positive integer: -12
Please enter a positive integer.
```